

MID TERM EXAMINATION

Subject: Linked Lists, Stacks, Queues

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 38 | Date: 11-03-2026

Total Questions: 10 | Answer all questions.

Section 1: MCQ (4 Questions - 8 Marks)

Q1. [2 marks] (Easy)

What is the primary characteristic that distinguishes a singly linked list node from an array element?

Options:

A) It stores only data. B) It stores data and a pointer to the next element. C) It stores data and a pointer to the previous element. D) It has a fixed memory location.

Q3. [2 marks] (Easy)

Which operation is used to add an element to the rear (end) of a queue?

Options:

A) pop B) push C) enqueue D) dequeue

Q7. [2 marks] (Easy)

In a singly linked list, which part of a node contains the address of the subsequent node?

Options:

A) Data field B) Value field C) Next pointer D) Previous pointer

Q10. [2 marks] (Medium)

Which data structure is explicitly designed to allow efficient insertion and deletion of elements from both its front and rear ends?

Options:

A) Stack B) Circular Queue C) Priority Queue D) Deque (Double-ended Queue)

Section 2: Short Answer (6 Questions - 30 Marks)

Q2. [5 marks] (Medium)

Explain the Last-In, First-Out (LIFO) principle as it applies to a stack data structure, and provide a real-world analogy.

Q4. [5 marks] (Hard)

Describe a scenario where a doubly linked list would be more advantageous to use than a singly linked list, explaining the specific benefit.

Q5. [5 marks] (Medium)

Describe how an array can be used to implement a stack, highlighting the role of the 'top' pointer/index in both push and pop operations.

Q6. [5 marks] (Medium)

Explain the primary difference between a linear queue and a circular queue regarding their handling of memory utilization and potential for false overflow.

Q8. [5 marks] (Hard)

Without using explicit recursion (but using a stack), outline the conceptual steps to check if a given string of parentheses (e.g., "(()())") is balanced.

Q9. [5 marks] (Hard)

Consider a scenario where tasks are added to a queue, but each task has a priority level. Briefly explain, conceptually, how a standard queue data structure could be adapted or what modifications would be needed to handle these prioritized tasks, without detailing specific code.

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